MD SHAH IMRAN SHOVON

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EDUCATION

Ph.D. in Electrical & Computer Engineering (Ongoing) January 2025 – Expected July 2029
University of Florida, Gainesville, Florida CGPA: 3.73/4.00
Relevant Coursework: Image Processing and Computer Vision, Fundamental Machine Learning, Semiconductor Packaging, Physical Assurance and Inspection of Electronics

B.Sc. in Electrical & Electronic Engineering July 2020
Shahjalal University of Science and Technology, Sylhet, Bangladesh
Relevant Coursework: Signals & Linear Systems, Digital Signal Processing, Numerical Analysis

PROFESSIONAL EXPERIENCE

PhD in ECE (Ongoing), University of Florida, GNV, FL, USA **Jan 2025 – Present**
Graduate Research Assistant

- Built a pinhole-based pipeline for counterfeit IC detection, from dataset curation to model evaluation.
- Co-developed WaveFormer for 3D medical image segmentation with large-scale multi-GPU training.
- Trained and applied X-ray laminography/tomography workflows (SIGRAY APEX Hybrid) for Defect and Failure analysis of Semiconductor devices.
- Contributed to dataset and model development by designing prompt schemas, preprocessing pipelines, and in-house data annotation tools; performed LLM and VLM fine-tuning with domain-specific datasets.

HowardMiller, Zeeland, USA **Oct 2021 – Dec 2024**
Full Stack Developer (Remote)

- Developed AI-driven deep learning strategies to optimize PPC budget allocation, reducing spend by 20% while maintaining the same sales volume (TensorFlow, Keras, Pandas, Scikit-Learn).
- Designed and optimized features, integrating third-party APIs using Flask, Django, Celery, Redis, Postgres, AWS, React.js, and related tools.
- Collaborated with stakeholders to gather and document user requirements, translating them into technical specifications and developing optimized features.

Gmonster.co, Balatonalmádi, Hungary **Sep 2018 – Oct 2021**
Project Lead (Remote)

- Oversaw the full development pipeline and built a Mass Email Marketing App with a subscription system from scratch.
- Engineered the system to send 10K+ emails per instance using separate proxy pipelines, leveraging concurrency and async processing for 20% higher efficiency than competitors.
- Achieved rapid adoption with 500+ users acquired within the first 2 months.

EnableGeek, Dhaka, Bangladesh **Jun 2022 – Dec 2024**
Founder

- Designed SEO pipeline, growing traffic to 500+ daily clicks within a year.
- Managed content and tech teams; published two books on Python and Java.

PROJECTS HIGHLIGHTS

Physical Inspection of Electronics: Counterfeit Detection | Research Project 2025 – Present

- Designed pinhole-image pipeline with automated defect ROI extraction for IC packages.
- Trained CNN/Transformer classifiers on in-house dataset for counterfeit detection with high precision/recall.
- Built multimodal optical-image dataset and VLM training setup for extended classification and explainability.

LLaMA Fine-Tuning with LoRA & QLoRA | Research Project 2025 – Present

- Contributed to physical-inspection dataset development (collection, labeling standards, QC).

- Set up reproducible training environments (Pipenv, Pyenv) on lab servers.
- Assisted in distributed training (torchrun) and optimized codebase (modularization, speedups, configs).
- Added pytest coverage and conducted code reviews for style and reliability.

Machine Learning | Artificial Intelligence | Computer Vision Projects 2017 – Present

- Analyzing the Consequences of High-Intensity Light Yolov5 and Strategies for Defending Against Adversarial Attacks (Yolo, Python) GitHub
- Custom Object Detection and Tracking using Darknet (Darknet-yolo, Python) GitHub, Youtube
- Dataset Creation for Training Custom Yolo Model (Yolo, Linux Shell Scripting) Drive

Academic Projects 2014 – 2020

Collaborated as a "Research Assistant" on three Official Projects supported by the UGC (University Grants Commission), leading the last two initiatives as Team Leader.

- Built an Autonomous Underwater Vehicle with navigation system (Arduino, Microcontroller).
- Developed an RFID-based automatic attendance system (Python, Sockets, SQL).
- Designed a UGV and Human Assistant system (Python, Raspberry Pi, Image Processing, LiDAR).

PUBLICATION

- Al, Zaman, M., Jawad, A., Santamaria-Pang, A., Lee, H. H., Tarapov, I., See, K., **Imran, M. S.**, Roy, A., Fallah, Y. P., Asadizanjani, N., & Forghani, R. (2025). **WaveFormer: A 3D Transformer with Wavelet-Driven Feature Representation for Efficient Medical Image Segmentation.** ArXiv.org. https://arxiv.org/abs/2503.23764 (MICCAI 2025)
- **Shah, M.**, Shajib Ghosh, Istiaq Firoz Shiam, Patrick Craig, Antika Roy, Pratyush Shukla, Varun Sai Vemuri, Hafez Bahrami, and Navid Asadizanjani (2026) **GUIDE-X: VLM-LLM-based AI assistant for artifact detection and operator guidance in x-ray inspection** (Accepted 2026 SPIE)
- Shajib Ghosh, Patrick Craig, Antika Roy, **Md Shah Imran Shovon**, Varun Sai Vemuri, John Allgair, Hafez Bahrami, and Navid Asadizanjani (2026) **AURA-PIE: an AI-enabled understanding and retrieval-augmented assistant for physical inspection of electronics** (Accepted 2026 SPIE)
- **PINCLASS: Pinhole-Centric Counterfeit Classification of IC Packages**. **Md Shah Imran Shovon**, Chih-Yun Pai, Shajib Ghosh, Chien-Chia Huang, Patrick J. Craig, Navid Asadizanjani. (PAINE 2025).
- **A Novel Framework for Identifying Counterfeit ICs via Pinhole Evaluation**. **Md Shah Imran Shovon**, Shajib Ghosh, Patrick J. Craig, Chien-Chia Huang, Chih-Yun Pai, Navid Asadizanjani. (RAPID 2025).
- **Deep Learning-Driven X-ray Analysis for High-Throughput Manufacturing**. Patrick J. Craig, Antika Roy, Shajib Ghosh, **Md Shah Imran Shovon**, Nitin Varshney, Navid Asadizanjani. (RAPID 2025).
- **Shah, M.** (2024). **Evaluating YOLO Object Detection Models Under Adverse Lighting Conditions.** Bulletin of the American Physical Society; American Physical Society. https://meetings.aps.org/Meeting/MAR24/Session/NN00.63
- **Shah, M.** (2024). **Evaluating and Enhancing Image Caption Generation: A Comparative Study of LSTM and GRU Models with Object and Feature Recognition Strategies.** Bulletin of the American Physical Society; American Physical Society. https://meetings.aps.org/Meeting/MAR24/Session/FF00.50
- For a complete list of publications, please visit my Google Scholar profile: **Link**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++, C, Matlab, Processing, Arduino, Flutter, Dart, Assembly Language (8086)
Libraries: PyTorch, TensorFlow, Keras, YOLO, NumPy, Scipy, Pandas, Selenium, Requests, Beautifulsoup4, Scrapy, Matplotlib, Seaborn, Bokeh, Django, Flask, React, Typescript,
Databases: SQL, Firebase, MongoDB, Pickle
Environments: Anaconda Environment, Jupyter Notebook, Pipenv

TRAINING & EXTRA CURRICULAR ACTIVITIES

- Led and participated in 12+ national robotics competitions (Line Follower, Maze/Grid Solver, Robo Fights) and electronics fairs, achieving 1st runner-up at Robomania, IUT (2015) with a multi-bot system (line following, obstacle avoidance, RC object carrier).

- Completed Top Up IT Java training by Ernst & Young (LICT project), certified by George Washington University.
- Completed AWS Cloud Practitioner course (2020), covering analytics, compute, serverless, containers, databases, and networking.